

Scala + Play + HMRC

Updated 12-Week Training Challenge and Canonical Daily Tracker

Version: 10 October 2026 | Goal: confidently read, modify, debug and build HMRC-style Scala/Play frontends

Confirmed completed	Current	Next
Week 1 Days 1-5; Week 2 Days 6-9	Week 2 Day 10 - Sealed Traits, Scala Enums and stronger domain modelling	Week 2 Day 11 - Integrated Week 2 assessment

Important correction: the original source plan fixed the weekly themes and 45-60 minute weekday format, but the early daily sequence was generated dynamically. This revision locks the daily sequence and makes the integrated assessment mandatory at the end of every week.

1. Training Contract

- **Duration:** 12 core weeks, Monday-Friday, approximately 45-60 minutes per session.
- **Week 2 repair:** one extra session is added because sealed traits/enums were missed earlier. This makes the programme 12 core weeks plus one repair session.
- **Progression rule:** never advance until the current exercise has been submitted and reviewed.
- **Assessment rule:** every week ends with an integrated assessment. A week is not complete until that assessment is reviewed.
- **Teaching style:** concise theory for new concepts, then a coding challenge, then review/refactoring.
- **Code examples:** no sample solution code unless explicitly requested. Hints come before solutions.
- **Learning style:** reuse earlier concepts naturally so knowledge remains cumulative. Use small unrelated examples and C# comparisons when helpful.

2. Session Format

Part	Typical time	Purpose
Theory / concept	10-15 min	Understand the new concept, why it exists, and how it maps to C# where useful.
Coding challenge	25-35 min	Apply the concept in a focused task that also reuses earlier learning.
Review / refactor	10 min	Correctness, Scala style, boundaries, simplification and readiness for the next topic.

3. 12-Week Roadmap

Week	Core focus	Mandatory weekly outcome
1	Scala fundamentals	Values/types, methods, expressions, if/else, Boolean logic, integrated assessment.
2	Data modelling + pattern matching	Case classes, companion objects, sealed traits, enums, pattern matching, guards, domain modelling, integrated assessment.
3	Collections	List/Seq, Set/Map, common operations, filtering/finding/grouping, immutable design, assessment.
4	Functional transformations	map, flatMap, collect, fold, foreach vs map, for-comprehensions, assessment.
5	Option / Either / error handling	Safe absence, domain errors, validation-style flows, assessment.
6	Future / async programming	Future composition, ExecutionContext, recovery, sequential/parallel reasoning, assessment.
7	Advanced Scala / application structure	Traits, packages, dependency boundaries, domain/service separation, immutability and assessment.
8	Play Framework fundamentals	Project structure, routes, controllers, Actions/Results, DI, config, Twirl, assessment.
9	Play forms and validation	Forms, mappings, validation, controllers, views, multi-page flow, assessment.
10	HMRC-style frontend patterns	HMRC conventions, UserAnswers/session state, navigation, connectors, config/toggles, assessment.
11	Testing and mocks	ScalaTest, Mockito, service/controller/async tests, boundaries, integration-style testing, assessment.
12	Final HMRC-style frontend project	Build and review a small Play frontend using the full learning stack.

4. Confirmed Progress and Week 2 Repair

Day	Topic	Status
1	Values and Types	Completed
2	Methods	Completed
3	Expressions	Completed
4	if/else and Boolean conditions	Completed
5	Week 1 integrated assessment	Completed
6	Case Classes and simple domain modelling	Completed
7	Objects and Companion Objects	Completed
8	Basic Pattern Matching	Completed
9	Pattern Matching with Case Classes and Guards	Completed
10	Sealed Traits + Scala Enums + stronger domain modelling	Current
11	Week 2 integrated assessment	Next

Why Day 11 exists: Week 2 originally progressed without covering sealed traits/enums. Day 10 repairs that missing concept; Day 11 then assesses the complete Week 2 scope before Week 3 begins.

5. Canonical Daily Plan - Week 3 to Week 6

Week 3 - Collections

Day	Topic	Learning target
Day 12	List and Seq	Create and use ordered immutable collections; indexing/head/tail where appropriate; reuse case classes.
Day 13	Set and Map	Uniqueness, key/value modelling, lookups and updates using immutable collections.
Day 14	Common operations	filter, find, exists and other common collection operations in realistic rules.
Day 15	Grouping, sorting and immutable design	groupBy/sort-style thinking; model collection data cleanly without mutation.
Day 16	Integrated collections assessment	Use List/Seq/Set/Map plus filtering/finding/grouping and prior domain concepts.

Week 4 - Functional transformations

Day	Topic	Learning target
Day 17	map and foreach vs map	Transform data; understand when foreach is side-effecting and map returns transformed data.
Day 18	flatMap and flatten	Transform nested collections and remove one level of nesting.
Day 19	collect and transformation pipelines	Selective transformation with pattern matching and multi-step pipelines.
Day 20	fold and collection for-comprehensions	Aggregate values and express chained collection transformations.
Day 21	Integrated functional assessment	Combine map/flatMap/collect/fold/for-comprehensions with earlier collection/domain skills.

Week 5 - Option, Either and error handling

Day	Topic	Learning target
Day 22	Option / Some / None	Represent missing values without null.
Day 23	Working safely with Option	map, flatMap, getOrElse, fold; avoid unsafe get and null.
Day 24	Either / Left / Right	Represent success or domain failure explicitly.
Day 25	Domain errors and validation flows	Model meaningful errors and chain validation-style logic.
Day 26	Integrated Option/Either assessment	Combine safe optional data, domain errors, pattern matching and earlier concepts.

Week 6 - Future and async programming

Day	Topic	Learning target
Day 27	Future basics + ExecutionContext	Understand asynchronous results and execution context.
Day 28	Future map and flatMap	Transform and chain asynchronous results.
Day 29	Future for-comprehensions	Compose dependent asynchronous operations.
Day 30	Recovery and async reasoning	Recover failures; reason about sequential vs parallel work.
Day 31	Integrated Future assessment	Compose async business logic with safe error handling and previous domain concepts.

6. Canonical Daily Plan - Week 7 to Week 9

Week 7 - Advanced Scala and application structure

Day	Topic	Learning target
Day 32	Traits and abstract classes	Use behavioural contracts appropriately and compare with C# interfaces/abstract classes.
Day 33	Packages and modules	Organise Scala application code into readable packages/modules.
Day 34	Dependency boundaries	Separate domain, service and infrastructure responsibilities.
Day 35	Immutability and pure functions	Improve code design, testability and readability using functional principles.
Day 36	Integrated structure assessment	Refactor a small domain application into clear layers and boundaries.

Week 8 - Play Framework fundamentals

Day	Topic	Learning target
Day 37	Play structure + request lifecycle	Understand app/conf/test layout and request-to-response flow.
Day 38	Routes, controllers, Actions and Results	Handle HTTP routes and produce Play Results.
Day 39	Dependency injection + configuration	Wire components and read environment/application configuration.
Day 40	Twirl views + simple end-to-end flow	Render a view through route -> controller -> result -> view.
Day 41	Integrated Play fundamentals assessment	Build a small end-to-end Play feature using the week concepts.

Week 9 - Play forms and validation

Day	Topic	Learning target
Day 42	Form and mappings	Define form models and mappings.
Day 43	Validation + binding/unbinding	Validate input and understand successful/failed binding.
Day 44	Controller form handling	GET/POST handling, errors, redisplay and successful submission.
Day 45	Views + simple multi-page flow	Connect forms, views and navigation across more than one page.
Day 46	Integrated forms/journey assessment	Build and validate a small Play form journey.

7. Canonical Daily Plan - Week 10 to Week 12

Week 10 - HMRC-style frontend patterns

Day	Topic	Learning target
Day 47	HMRC frontend/bootstrap conventions	Recognise common HMRC frontend project conventions and dependencies.
Day 48	UserAnswers and session state	Model journey data/state and understand persistence/session patterns.
Day 49	Navigation and journey flow	Choose next pages based on answers and business rules.
Day 50	Connectors, config/toggles and layering	Combine connector/service/controller/model/config responsibilities.
Day 51	Integrated HMRC-style journey assessment	Implement/reason through a small HMRC-style journey using the week concepts.

Week 11 - Testing and mocks

Day	Topic	Learning target
Day 52	ScalaTest fundamentals	Structure tests, assertions and test data.
Day 53	Mockito, mocks and stubs	Isolate dependencies and verify interactions.
Day 54	Service/controller and async tests	Test business logic, controllers and Future-based behaviour.
Day 55	Boundary/error-path and integration-style tests	Cover failures, edge cases and HMRC-style test organisation.
Day 56	Integrated testing assessment	Test a feature across multiple layers with meaningful coverage.

Week 12 - Final HMRC-style frontend project

Day	Topic	Learning target
Day 57	Scope, design and models	Define the mini-project, domain model and journey.
Day 58	Forms, controllers and views	Implement user-facing flow.
Day 59	Services, connectors, config and async flow	Add application/service/integration behaviour.
Day 60	Tests and refactoring	Add tests, simplify code and address design issues.
Day 61	Final project review and assessment	Review correctness, Scala/Play style, HMRC relevance, testing and remaining gaps.

8. Assessment and Tracking Rules

- **Lesson completion:** a lesson is completed only after the exercise is submitted and reviewed.
- **Week completion:** a week is completed only after its integrated assessment is submitted and reviewed.
- **No silent advancement:** if a task is unfinished, the next scheduled session repeats the current checkpoint rather than generating a new topic.
- **Assessment scope:** each integrated assessment tests the current week plus selected earlier concepts so retention is cumulative.
- **Progress header:** every lesson must show Completed / Current / Next and the current week checklist.
- **Repository:** continue using the same scala-play-training repository; fundamentals first, then the Play/HMRC training app as the plan reaches Play.

9. Current Checkpoint

Item	Current state
Last reviewed lesson	Week 2 Day 9 - Pattern Matching with Case Classes and Guards
Current lesson	Week 2 Day 10 - Sealed Traits + Scala Enums + stronger domain modelling
Next lesson	Week 2 Day 11 - Integrated Week 2 assessment
Next week	Week 3 - Collections, starting with List and Seq

Source alignment: this document preserves the agreed 12-week weekly focus and 45-60 minute challenge-first learning format, while adding the fixed daily sequence and assessment gates needed to prevent future drift.